



Operation optimisation of micro pumped storage for decarbonisation of water and electricity distribution network in urban areas

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ABSTRACT

The rapid growth in renewable energy generation has heightened the need for enhanced energy storage capacity. Micro Pumped Storage (MPS) offers a dual-function alternative to conventional batteries by integrating energy storage with non-potable water supply (NPWS) in high-rise buildings. This study hypothesises that jointly optimising MPS operation for emissions and water reuse can improve both environmental and economic performance. A multi-objective genetic algorithm (GA) is developed to optimise monthly scheduling under time-varying electricity prices and emission factors, incorporating penalty-based constraints to ensure water supply reliability. Our findings indicate that optimising MPS schedule can reduce operational emissions, with reductions of 4 % to 25 % in ES mode, and 25 % to 43 % in NPWS mode. Additionally, the NPWS mode reduces potable water consumption by up to 3000 m³ annually, which translates to saving 54 MWh of energy and 44 tonnes of CO₂ within Melbourne's water distribution network. These environmental and economic benefits contribute to a reduction in the payback period of MPS installations by up to 20 years, particularly in NPWS mode. Compared to equivalent-capacity lithium-ion batteries, MPS systems offer shorter payback periods and additional sustainability benefits. Furthermore, MPS provides a viable pathway to secure sustainability certification such as Green Star, and support urban decarbonisation through integrated water-energy optimisation. Practical challenges, however, must be addressed for wider adoption.

1. Introduction

The growth in renewable energy generation has intensified the need for greater energy storage capacity load demand management. Pumped storage, one of the most conventional energy storage methods (Blakers et al., 2021), alongside utility-scale batteries, plays a pivotal role in load demand management. Although micro pumped storage (MPS) can outperform batteries in specific urban configurations (Boroomandnia et al., 2024), its economic performance is significantly influenced by the operational schedule. Therefore, optimising MPS scheduling is essential to maximise profitability and support electricity decarbonisation. Various optimisation models have been applied to energy storage systems to improve economic performance. Zhang and Zhang (2013, 2014) used a Genetic Algorithm (GA) to optimise MPS operations for energy arbitrage in a high-rise building, while Kusakana (Kusakana, 2018) applied linear programming (LP) to minimise the levelised cost of storage (LCOS) in irrigation systems powered by an elevated water tank.

Georgiou et al. (Georgiou et al., 2019) optimised battery dispatch in PV-equipped buildings, and Akkaş and Çam (AKKAŞ & Ertugrul, 2020) developed operational strategy for a virtual power plant (VPP) with pumped hydro storage. Similarly, optimisation of pump operations has been explored through economic model predictive control framework in real-time water distribution systems (Wang et al., 2021), and through machine learning-based energy management strategies for peak shaving (Sarmas et al., 2022), demonstrating the potential of optimal scheduling to enhance energy efficiency without additional costs. Zhou et al. (Zhou et al., 2022), Pandžić et al. (Pandžić et al., 2013) and Vecchi et al. (Vecchi et al., 2020) further applied a stochastic mixed-integer linear programming (MILP) model to enhance energy arbitrage profits through optimal dispatch profiles. While, these studies highlight the effectiveness of various optimisation models in improving financial performance, they often overlook emission reduction and the influence of time-varying electricity tariffs. To address some of these limitations, Stoppato et al. (Stoppato et al., 2016; Stoppato et al., 2014) optimised a hybrid cogeneration system with batteries, a water reservoir, and

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Nomenclature			
<i>Symbols</i>			
<i>Arb</i>	energy arbitrage, A\$	<i>CAPEX</i>	capital expenditure
<i>d</i>	penstock diameter, m	<i>NPV</i>	net present value
<i>E</i>	electrical energy, kWh	<i>EMPC</i>	economic model predictive control
<i>g</i>	gravitational acceleration, m/s ²	<i>NPWS</i>	non-potable water supply
<i>H</i>	head, m	<i>ES</i>	energy storage
<i>m</i>	penalty factor operational const	<i>OE</i>	operational emissions
<i>n</i>	penalty factor operational emission	<i>GA</i>	genetic algorithm
<i>P</i>	power, kW	<i>OPEX</i>	operating expenses
<i>Q</i>	water flow, m ³ /s	<i>GBCA</i>	green building council of Australia
<i>t</i>	timeslot identifier	<i>PAT</i>	pump as turbine
<i>Vol</i>	water storages volume, m ³	<i>GR</i>	green roof
<i>x</i>	time-varied electricity spot price, A\$/kWh	<i>PHS</i>	pumped hydro storage
<i>y</i>	time-varied grid emission factor, ton CO ₂ / kWh	<i>HI</i>	hypervolume indicator
<i>Greek symbols</i>		<i>PP</i>	prosumers power
Δt	timeslot, h	<i>LCOS</i>	levelised cost of storage
γ	specific gravity of water, kN/m ³	<i>PSO</i>	particle swarm optimisation
η	pump/turbine efficiency	<i>LEE</i>	levelised embodied emissions
<i>Subscripts</i>		<i>RI</i>	renewable Intensive
<i>P</i>	pump	<i>LEED</i>	leadership in energy and environmental design
<i>OE</i>	operational emissions	<i>S</i>	nearby stream
<i>OC</i>	operational cost	<i>Li-ion</i>	lithium-ion batteries
<i>T</i>	turbine	<i>T</i>	modular tank
<i>Abbreviations</i>		<i>LM</i>	linear programming
<i>AEMO</i>	Australian energy market operator	<i>US</i>	upper storage
<i>NABERS</i>	national Australian built environment rating system	<i>LS</i>	lower storage
		<i>VPP</i>	virtual power plant
		<i>MILP</i>	mixed integer Linear programming
		<i>WEN</i>	water-energy nexus
		<i>MPS</i>	micro pumped-storage
		<i>WSUD</i>	water sensitive urban design

thermal storage using Particle Swarm Optimisation (PSO), though they did not account time-varied greenhouse gas emissions. In contrast, Segurado et al. (Segurado et al., 2016) and Bhimaraju et al. (Bhimaraju, Mahesh & Nirbheram, 2023) employed a multi-objective optimisation approach to reduce both cost and CO₂ emissions. Likewise, Yang et al. optimised electric buses operation participating in carbon market to reduce energy costs and emissions (Yang et al., 2021). However, none of these studies accounted for the temporal variability of the grid's emission factor or explored synergy between the water and energy system in urban scale — both of which are addressed in this study.

Focusing on real-time emission factors, Aryai and Goldsworthy (Aryai & Goldsworthy, 2022) created a model for the Australian National Energy Market using a power-flow tracing approach to quantify the grid's carbon emissions intensity. Their framework, applied to residential battery storage units, demonstrated how emission reduction objectives can influence the financial performance of storage systems. Yang et al. (Yang & al., 2024) also proposed an urban scale decarbonisation framework, inspired by energy trade in liberalised markets to accelerate net-zero goals. Inspired by these approaches, our study explores the impact of emission minimisation on the energy arbitrage potential of MPS in high-rise buildings. MPS also offers the additional advantage of supplying water, presenting a unique dual-purpose benefit. Valencia et al. (Valencia, Hossain & Chang, 2022) showed that integrated management of food, energy, water, and waste at the community scale enhances emission reduction through synergistic infrastructure optimisation. Silva and Hendrick examined the economic feasibility of MPS in Arras, France, but called for further investigation into its application for water supply, and overall economic performance (de Oliveira e Silva & Hendrick, 2016). Similarly, Walker and Duquette (Walker & Duquette, 2022) assessed MPS and gravity module systems in Canada, focusing solely on economic factors without addressing operational emissions or water supply feasibility. While Chiu et al. (Chiu,

Liaw & Chen, 2009) proved that decentralised water systems can reduce the energy intensity of urban water networks in hilly regions of Taiwan, the potential similar savings in flatter urban environments remains untouched. Although Zhang et al. (Zhang et al., 2009) and Cook et al. (Cook, Sharma & Gurung, 2014) demonstrated the financial benefits and minimal energy requirements of alternative water use in Australian cities, they did not evaluate the associated emission reductions at urban scale. Addressing these gaps, this study evaluates the combined impact of non-potable water supply and energy arbitrage on the economic feasibility and operational emissions of MPS systems at building level, and extends the analysis to city-wide water supply network.

The Water-Energy Nexus (WEN) evaluation in this study opens new avenues for sustainability assessment, as water and energy are key elements in sustainability rating systems such as Green Star (Hafez et al., 2023), Leadership in Energy and Environmental Design (LEED), the National Australian Built Environment Rating System (NABERS), and similar tools. Sustainable building practices have gained global significance among researchers and construction industry authorities. Alawneh et al. (Alawneh et al., 2018) investigated the relationship between water efficiency and energy and atmosphere in LEED certified buildings in Jordan, linking them to Sustainable Development Goals. Asdrubali et al. (Asdrubali et al., 2015) compared LEED and ITACA (Italian certification system), by applying both to residential buildings in Italy. From a healthy environment perspective, Roumi et al. (Roumi et al., 2021) and Residovic (Residovic, 2017) studied the correlation between NABERS Energy ratings and indoor environmental quality, exploring impacts on productivity and well-being. Similarly, Gui and Gou (Gui & Gou, 2020) assessed how green certification levels influence post-occupancy performance in office buildings. From an economic viewpoint, green building investments have shown favourable returns in the real estate market (Mansour & Radford, 2016). Based on these findings, evaluating the MPS unit through sustainability rating tools

provides practical insights for building sector authorities and address a gap by investigating its impact on Green Star rating.

In summary, while existing research has advanced the optimisation of energy storage systems and urban water-energy management, several critical gaps remain. As summarised in Table 1, prior studies have largely focused on economic performance without assessing how emission reduction objectives affect energy arbitrage. Moreover, synergies between micro pumped storage systems and non-potable water supply have been overlooked. The broader impacts of integrating MPS into city-scale water-energy networks, and its contribution to sustainability ratings such as Green Star, have also not been evaluated. This study addresses these gaps by developing an integrated optimisation model that minimises operational emissions, accounting for time-varying emission factors, while enhancing the economic viability of MPS, quantifying its dual benefits for energy and water supply, and assessing its impact on building sustainability certification. In doing so, it offers a novel approach to advancing urban decarbonisation through the combined lens of operational efficiency, water-energy nexus optimisation, and sustainability benchmarking.

2. Research methods

2.1. Pumped-storage operation modelling approach

A decision-making unit is crucial for selecting the best design among optimal configurations that minimise levelised embodied emissions (LEE) and LCOS, outperforming lithium-ion batteries. Studies comparing energy storage configurations highlight weighted factors in decision-making. Babatunde et al. found that technical criteria contributed 63 %, economic criteria 26.6 %, and environmental criteria 10.4 % (Babatunde, Munda & Hamam, 2022). Similarly, Qie et al. used a multi-criteria decision-making framework showing a weight ratio of economic factors to environmental factors of 3 to 1 (Qie et al., 2021). Garcia-Bernabeu et al. (Garcia-Bernabeu, Mayor-Vitoria, Bravo & Pla-Santamaria, 2019) combined the VIKOR and AHP methods revealing a 3 to 1 ratio between NPV and avoided carbon emissions. Therefore, in this study, a rational decision-making model was adopted, reflecting a similar weight ratio of 3 to 1 for LCOS to LEE, as supported by literature on energy storage systems. The selected configuration for hydraulic

modelling was determined by minimising the weighted sum of LCOS and LEE, ensuring an optimal balance between cost efficiency and environmental impact.

The best design characteristics, including water storage volume and type, MPS flow rate, penstock diameter, and pump-as-turbine features, are inputs for the hydraulic model. This hydraulic model, developed in Python, simulates MPS operation by considering minor and major losses from water flow in pipes, water storage constraints, and a diurnal non-potable water use profile for toilet flush. The spatial scale of the developed optimisation framework is a single high-rise building, and the temporal resolution is 30-minute intervals to generate a monthly optimal operational schedule. Fig. 1 illustrates the integrated modelling and optimisation framework developed in this study. The framework begins with input data, including operational modes for 30-minute time slots, wholesale electricity tariffs, 30-minute grid emission factors, diurnal water consumption profiles in buildings, and MPS configurations with optimal designs filtered by Decision-Making Unit. These inputs are processed by the MPS Hydraulic Modelling Unit, which simulates system behaviour based on flow rates, water storage constraints, and energy conversion parameters. The outputs are then evaluated in the Eco-economic Analysis Unit, which calculates operational emissions, energy arbitrage, and water-energy savings. The GA generates candidate operational schedules and iteratively improves them through selection, crossover, and mutation processes. The GA seeks schedules that maximise energy arbitrage and minimise operational emissions, while ensuring compliance with storage constraints. As part of this process, wholesale electricity tariff data and grid emission factor data from 2021 were used to identify the optimum monthly operation schedule.

In the hydraulic modelling unit, the Darcy-Weisbach Equation is employed to calculate major head losses, while minor head loss coefficients are determined based on the types of bends, elbows, joints, and valves in the system. These head losses are summed and adjusted for the building height, adding for pumping mode and subtracting for releasing mode. Given the negligible height of the water storage compared to the building height, variations in water level have an insignificant impact on the total hydraulic head, thus minimally affecting pump/turbine performance and energy generation. Subsequently, pump power and turbine power are calculated using Eqs. (1) and (2).

Table 1

Comparison of objectives and scope between existing literature and the current study.

Reference	System Type	Research Scale	Optimisation Method	Objective(s)	Time-Varying Emission Factor	WEN	Sustainability Certification
(Zhang & Zhang, 2013; Zhang & Zhang, 2014)	MPS	Building	GA	Arbitrage	✗	✗	✗
(Kusakana, 2018)	PV + PHS	Building	LP	LCOS minimisation	✗	Partial	✗
(Georgiou et al., 2019)	Battery	Building	GA + LP	Arbitrage	✗	✗	✗
(AKKAŞ & Ertugrul, 2020)	PHS in VPP	System	MILP	Arbitrage	✗	✗	✗
(Wang et al., 2021)	Water distribution	Urban	EMPC	Energy cost	✗	Partial	✗
(Sarmas et al., 2022)	Water pumps	Urban	ML	Peak shaving	✗	✗	✗
(Zhou et al., 2022; Pandžić et al., 2013; Vecchi et al., 2020)	PHS in VPP	System	MILP, stochastic	Arbitrage	✗	✗	✗
(Stoppato et al., 2016; Stoppato et al., 2014)	Hybrid cogeneration + PHS	Building	PSO	Cost Emissions	✗	Partial	✗
(Segurado et al., 2016)	Wind + Desal + PHS	System	Direct multi-search	Cost Emissions	✗	✓	✗
(Bhimaraju, Mahesh & Nirbheram, 2023)	Hybrid cogeneration + PHS	System	Multiple Algorithms	Cost Emissions	✗	✗	✗
(Aryai & Goldsworthy, 2022)	Battery	Building	Emissions control	Arbitrage Emissions	✓	✗	✗
(de Oliveira e Silva & Hendrick, 2016)	MPS	Building	–	Feasibility	✗	✓	✗
(T. Walker & Duquette, 2022)	MPS and gravity	Building	Techno-economic analysis	Cost	✗	✗	✗
(Zhang et al., 2009; Cook, Sharma & Gurung, 2014)	Alternative water reuse	Building	Scenario analysis	Cost	✗	Partial	✗
This study	MPS	Building + Urban	GA	Arbitrage Emissions	✓	✓	✓

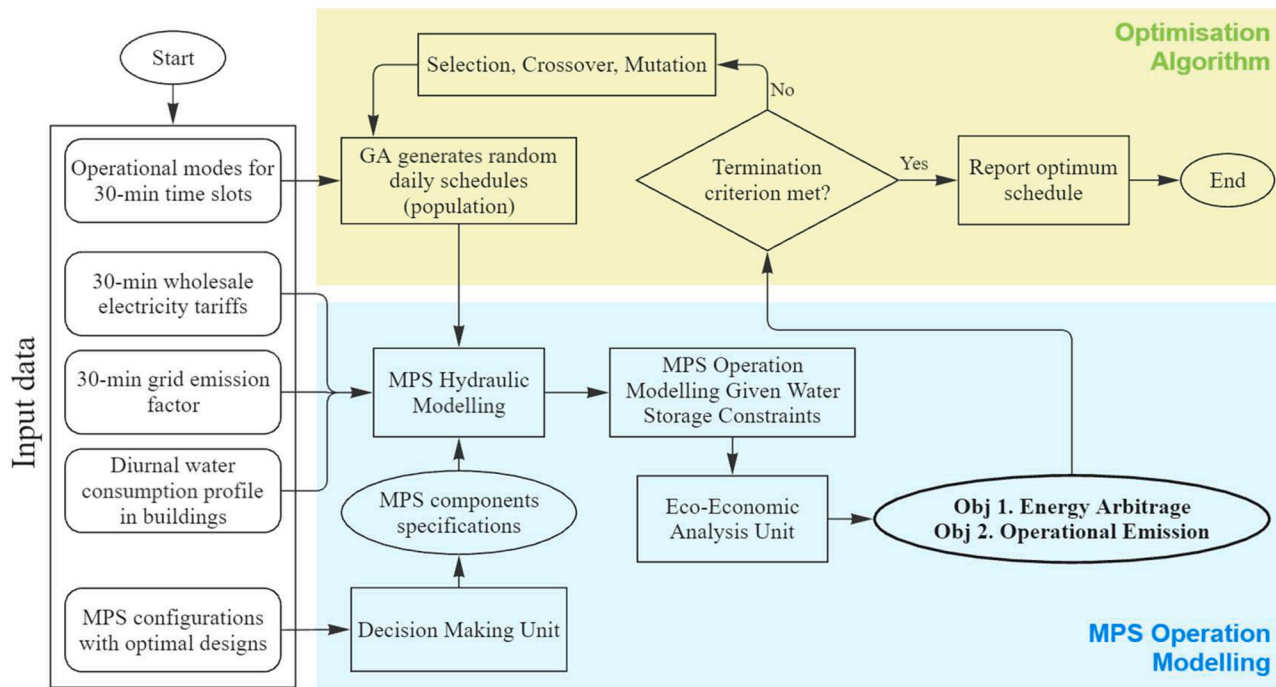


Fig. 1. Flowchart of the applied modelling-optimisation methodology.

$$P_T = Q_T \cdot H_T \cdot \gamma \cdot \eta \quad (1)$$

$$P_P = Q_P \cdot H_P \cdot \gamma / \eta \quad (2)$$

where P is power in kW, P and T subscripts stand for the pump and turbine, respectively, H is the total hydraulic head in m, Q is the water flow in m^3/s , γ is the specific gravity of water, which accounts for 9.81 kN/m^3 , and η is the machine efficiency.

Another condition that governs MPS operation is the water level status in the upper storage (US) and lower storage (LS). At the model's starting point, both water storages, which have identical volumes, are assumed to be half full, with 4 % of each storage's total volume designated as dead storage. As the MPS operates, the water levels in the storage will vary based on flow rate and runtime duration. In Energy Storage (ES) mode, the MPS follows one of three operational modes in each 30-minute time slot: consuming energy to pump water, generating energy via the turbine, or shutting down. In Non-Potable Water Supply (NPWS) mode, an additional operating state is introduced to supply base non-potable water demand in residential building from the US via gravity feed. The mathematical equations for the mentioned operating states are listed in Table 2. Equations (3–1), (4–1), (5–1), and (6–1) calculate the energy cost (A\$), assuming positive values when energy is consumed and negative values when energy is generated or saved. Equations (3–2), (4–2), (5–2), and (6–2) follow the same logic for calculating operational emissions (ton CO_2). Penalties in both groups of equations account for penalties due to constraint violations. These penalties are applied to ensure the system adheres to constraints, which maintains appropriate water levels in storages in this case. This method, known as the “penalty function method,” helps eliminate technically impossible solutions and ensures that the optimisation process focuses on feasible and practical solutions (Cimorelli, D’Aniello & Cozzolino, 2020).

In Table 2, Vol is the volume of water storage at time step t . The US and LS subscripts stand for the upper storage and lower storage, respectively. Δt is the operational time slot in seconds. x represents the spot price of electricity at Δt in A\$/ kWh, y represents grid emission factor at Δt in ton CO_2/kWh . m and n are penalty factors used in the penalty function method, and $npwd$ is non-potable water demand at time

Table 2

Mathematical equations representing MPS operation.

Operating state	Equations
Pump On	- if $Vol_{LS_t} < 0$ (3–1)
	Penalty $_{OCt} = m \cdot Vol_{LS_t} $ (3–2)
	Penalty $_{OEt} = n \cdot Vol_{LS_t} $ (3–3)
	- else if $Vol_{US_t} > Vol_{US_{max}}$: (3–4)
	Penalty $_{Arb_t} = m \cdot Vol_{US_t} - Vol_{US_{max}} $ (3–5)
	Penalty $_{OEt} = n \cdot Vol_{US_t} - Vol_{US_{max}} $ (3–6)
	- else: (3–7)
	Penalty $_{OCt} = 0$ (3–8)
	Penalty $_{OEt} = 0$ (3–9)
	- else: (3–10)
Turbine On	- if $Vol_{US_t} < 0$: (4–1)
	Penalty $_{OCt} = m \cdot Vol_{US_t} $ (4–2)
	Penalty $_{OEt} = n \cdot Vol_{US_t} $ (4–3)
	- else if $Vol_{LS_t} > Vol_{LS_{max}}$: (4–4)
	Penalty $_{OCt} = m \cdot Vol_{LS_t} - Vol_{LS_{max}} $ (4–5)
	Penalty $_{OEt} = n \cdot Vol_{LS_t} - Vol_{LS_{max}} $ (4–6)
	- else: (4–7)
	Penalty $_{OCt} = 0$ (4–8)
	Penalty $_{OEt} = 0$ (4–9)
	- else: (4–10)
Non-Potable Water Supply	- if $Vol_{US_t} < 0$: (5–1)
	Penalty $_{OCt} = m \cdot Vol_{US_t} $ (5–2)
	Penalty $_{OEt} = n \cdot Vol_{US_t} $ (5–3)
	- else: (5–4)
	Operational Cost $_t = 0$ (5–5)
	Operational Emission $_t = 0$ (5–6)
	Penalty $_{OCt} = 0$ (5–7)
	Penalty $_{OEt} = 0$ (5–8)
	- else: (6–1)
	Operational Cost $_t = 0$ (6–2)
MPS Off	Operational Emission $_t = 0$ (6–3)
	Penalty $_{OCt} = 0$ (6–4)
	Penalty $_{OEt} = 0$ (6–5)
	- else: (6–6)

step t in m^3 . Penalty factors vary for each engineering question and are fixed values defined according to the objective function values. In this research, m and n are considered to be 40 and 400, respectively.

Arbitrage is the profit from energy price differences. To calculate the energy arbitrage and operational emissions for a one-month period, Eqs. (7) and (8) are used. The negative sign in Eq. (7) is due to the fact that costs are considered positive, and revenues are considered negative. The variable dd in these two equations represents the number of days in each month and varies between 28, 30, and 31 based on the month the model is running for.

$$\text{Arbitrage} = - \sum_{t=1}^{48 \times dd} \text{Operational Cost}_t \quad (7)$$

$$\text{OE} = \sum_{t=1}^{48 \times dd} \text{Operational Emission}_t \quad (8)$$

Apart from arbitrage and operational emissions, which are the objective functions of the model, two more parameters are calculated in the Eco-economic Analysis Unit: saved energy within the water distribution network (kWh) and the payback period (year). If a nearby stream is used to run the MPS and supply non-potable water demand, less water will be supplied from the distribution network to the building node. This will result in energy savings within the water distribution network, which is calculated using Eq. (9). The payback period, as depicted in Eq. (10), is the amount of time it takes to recover an initial investment through the cash flows it generates. This parameter is used to determine which scenarios are more economically attractive.

$$\text{Saved Energy}_{DN} = k \cdot \sum_{\text{month}=1}^{12} \sum_{t=1}^{48 \times dd} npwd_t \quad (9)$$

$$\text{Payback Period} = \frac{\text{CAPEX}}{\sum_{\text{month}=1}^{12} \sum_{t=1}^{48 \times dd} \text{Arbitrage}} \quad (10)$$

where k is the energy intensity of the water distribution network in kWh/m^3 , CAPEX is capital expenditures in A\$. The energy intensity value reflects the energy required to extract, treat, and distribute water across the urban network, ensuring a reliable water supply for residential and commercial use.

2.2. Optimisation problem formulation

Evolutionary and metaheuristic algorithms surpass traditional optimisation methods in effectively addressing nonlinear and complex problems. In this study, the power terms of the pump and turbine in energy arbitrage, which depend on head and flow rate, introduce non-linearity. Evolutionary algorithms are particularly efficient for solving real-life problems like energy storage systems (Brebán, 2016), involving multiple continuous and integer decision variables. It has been demonstrated that as optimisation problems become more complex, computational time and memory usage increase exponentially for classical optimisation algorithms. However, genetic algorithms (GA) handle increased complexity more efficiently, with only a steady, linear rise in time and memory usage (East, 1994). This study employs Pymoo, a Python-based framework, to conduct a GA aimed at maximising the energy arbitrage and minimising operational emissions under various scenarios. In this optimisation problem, the main decision variables are the operational modes of the MPS system across 30-minute time intervals, represented as pumping, turbine generation, non-potable water supply, or off states. Key parameters include electricity prices, grid emission factors, pump and turbine efficiencies, water demand profiles, and storage tank constraints. The optimisation is driven by two primary objectives: maximising financial returns through energy arbitrage and minimising operational carbon emissions. Additionally, the model incorporates penalties to ensure compliance with storage constraints,

thereby maintaining non-potable water supply alongside energy management. These penalties were iteratively calibrated ($m = 40$, $n = 400$) to reflect the relative scale of the two objective functions and ensure that infeasible solutions are effectively excluded from evolution. The objective functions are defined as in Eqs. (11) and (12).

$$\text{fitness function1} = \max (\text{Arbitrage}) \quad (11)$$

$$\text{fitness function 2} = \min (\text{OE}) \quad (12)$$

The decision variables, which include four operating states (Fig. 2), influence both energy arbitrage and OE. Each solution is represented by a chromosome that comprises up to 48 decision variables, corresponding to the 48 time slots in a day. Further details on the GA parameters and convergence analysis are provided in Section 3.4.

3. Case study

3.1. MPS attributes

MPS units, akin to their larger counterparts, employ two water reservoirs: an US and a LS. The capacities of these storages are limited by rooftop load regulations, which require the distributed load on special-purpose roofs to range between 2.5 to 5 kPa. Besides modular tanks, alternative configurations for both US and LS include multilayer green roofs, blue roofs, and natural or artificial streams. In this study, we assume a practical maximum distance of 20 m from buildings to streams, enabling the use of these streams as LS. This assumption is based on the average proximity of existing high-rises to nearby streams and aims to minimise urban infrastructure disruption due to the lack of established guidelines. This approach was similarly applied in previous research that investigated the optimal design of MPS (Boroomandnia et al., 2024), striking a balance between feasibility and minimal urban impact derived from observed evidence.

The scenarios for US and LS and the range of building heights in the study area were coded as "upper storage type/lower storage type-height class" for assessment. This method resulted in the creation of 30 scenarios, considering three options for US and two configurations for LS across five building height classes. However, only scenarios outperforming Li-ion batteries in terms of LEE and LCOS were selected. Detailed explanations regarding LEE and LCOS calculations for each scenario and comparisons with batteries are available in our previous publication in the Journal of Energy Conversion and Management (Boroomandnia et al., 2024). Based on the results from that paper, configurations with blue roofs cannot compete with Li-ion batteries, making them not worth optimising for operation. Therefore, in this study, the potential US configurations include Modular Tanks (T) and Green Roofs (GR), while the LS options consist of Modular Tanks and Nearby Streams (S). Buildings were categorised into five height classes in meters: (50–100], (100–150], (150–200], (200–250], and (250–300]. However, only some of these building height classes outperform batteries. Operation of all configurations follows Eqs. (3) to (6), except for ES mode where Eq. (5) will be omitted. If a nearby stream is used as the LS, all terms and penalties relevant to the LS will also be omitted.

The design optimisation generates a Pareto front for each scenario, resulting in multiple cases for the MPS, each with a unique design featuring specific energy storage capacity and distinct characteristics regarding the pump as turbine (PAT) type, penstock diameter, charge/discharge flow rate, and water storage volume. To select the best design for each scenario and optimise its operation schedule, a rational decision-making model was adopted, reflecting a weight ratio of 3 to 1 for LCOS to LEE. The best designs selected for operation optimisation, are illustrated by stars in Fig. 3, while the other optimal designs are depicted as dots. Fig. 3 highlights 17 techno-economically feasible cases, providing a valuable basis for further analysis. The technical attributes of each design are listed in Table 3. The maximum storage volume and flow rate introduced to the model are 1000 m^3 and 125 l/s , respectively.

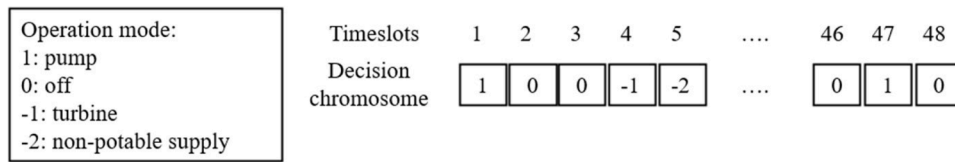


Fig. 2. Decision variables and chromosomes used in GA optimisation.

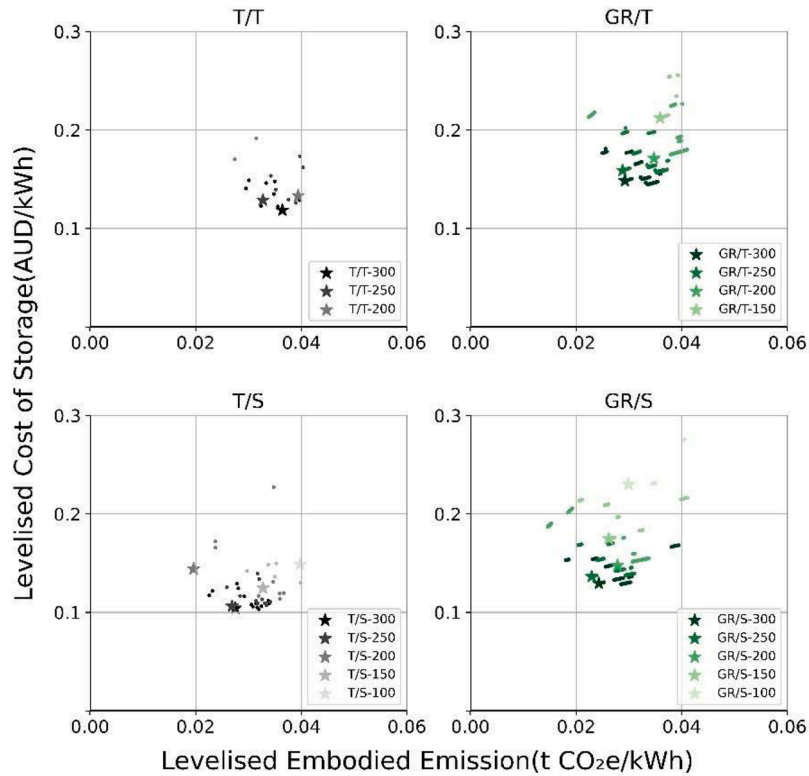


Fig. 3. Filtered cases with optimal designs for operation optimisation.

Table 3

Technical attributes of filtered cases for optimal operation.

MPS configuration	Storage volume (m ³)	Flow rate (L/s)	Penstock diameter (mm)
GR/S-100	704.2	90	200
GR/S-150	782.4	100	200
GR/S-200	704.2	90	200
GR/S-250	782.8	100	200
GR/S-300	783.5	100	200
GR/T-150	782.7	100	200
GR/T-200	704.2	90	200
GR/T-250	782.2	100	200
GR/T-300	781.4	100	200
T/S-100	724.0	90	200
T/S-150	779.1	99	200
T/S-200	957.6	81	180
T/S-250	791.3	100	200
T/S-300	784.3	100	200
T/T-200	726.3	90	200
T/T-250	783.1	100	200
T/T-300	975.6	124	225

According to Table 3, it is evident that the maximum flow rate and storage volume are not necessarily the best designs with the lowest LEE and LCOS.

In this study, Melbourne was chosen as the focus due to its substantial dependence on coal-fired power generation and its ongoing shift

toward renewable energy sources, underscoring the critical need for energy storage solutions. Additionally, Melbourne is home to many high-rise buildings, making it an ideal location to explore the integration of energy storage systems in urban settings. The original building dataset, including heights and rooftop areas, was sourced from the publicly available City of Melbourne data portal. In our previous studies (Boroomandnia et al., 2024; Boroomandnia et al., 2024), this dataset was analysed and filtered using a multi-objective optimisation framework to select only those buildings with optimal MPS designs that outperform Li-ion batteries in terms of economic and environmental performance. Therefore, the buildings analysed in this study represent a filtered subset of the metropolitan stock. The distribution of candidate buildings is shown in Fig. 4. When using nearby streams to run the MPS under NPWS mode, energy savings within the water distribution network are achieved as water demand is reduced. The energy intensity of the water supply network in Melbourne is approximately 1.74 kWh/m³ (Yan et al., 2024).

To calculate arbitrage based on the optimal monthly schedule, spot price electricity data are essential. The Australian Energy Market Operator (AEMO) has been providing 30-minute interval electricity tariff data since 1999 for five Australian states (AEMO, 2022). Prosumers can access real-time wholesale electricity tariffs and participate in the energy market through specific retailers in Australia, such as Amber Electric. In this study, wholesale electricity spot prices from 2021 are used because the 2021 data formed the basis for design optimisation

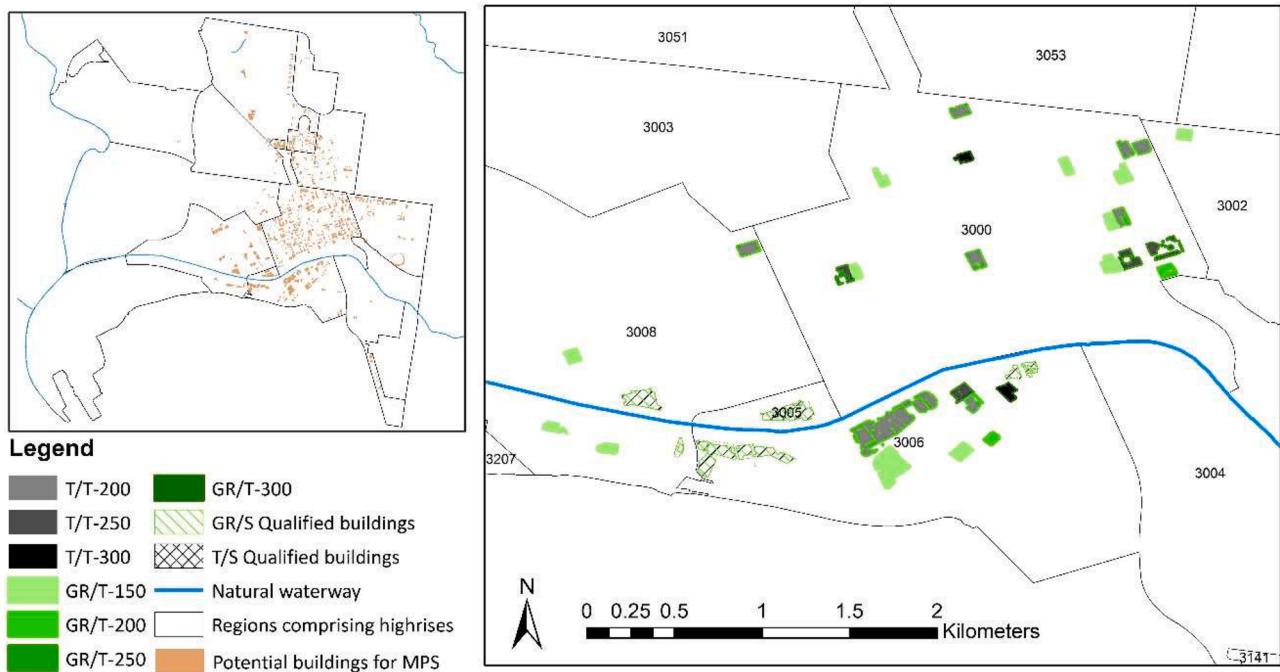


Fig. 4. Distribution of buildings selected for MPS operational optimisation based on economic and environmental performance in Melbourne (Boroomandnia et al., 2024). Electricity tariff and emission factor data.

and economic assessment of various MPS configurations in our previous research, which serves as the base case scenario herein Boroomandnia et al. (2024). To ensure consistency when comparing the base case with multi-objective optimisation scenarios that include operational emissions and arbitrage, the input data from 2021 are maintained.

Regarding real-time emission factors, Aryai and Goldsworthy developed a model for the Australian National Energy Market using a power-flow tracing approach to quantify the carbon emissions intensity of the electricity grid (Aryai & Goldsworthy, 2022). Their study revealed that the average and peak emissions intensities during low electricity tariff periods are equal to or slightly higher than those during high tariff periods. Moreover, they found no significant correlation between the 30-minute wholesale electricity price and the emissions intensity of electricity in 2021. Based on their model, which provides inputs for our analysis, we investigated how controlling an MPS unit for maximum arbitrage could also help minimise greenhouse gas emissions.

3.2. Water consumption data

In NPWS mode, alongside energy arbitrage, the goal of optimal operation is to reduce energy costs for water supply by shifting pump operation to times with lower electricity tariffs. The diurnal water consumption profile in Victoria in residential buildings, sourced from eWater toolkit resources, is used to achieve this (eWater, 2019). The eWater Group develops modelling tools to enhance water management. According to a recent Melbourne Water Outlook report, average per capita water consumption in Melbourne is 159 Liters per day (Water, 2022). To determine toilet flush demand per 30-minute timeslot, parameters such as the number of occupants and the percentage of total water use for toilet flushing are needed. Walker and Duquette's study on energy generation from wastewater in high-rise buildings assumes 100 occupants per level (Walker & Duquette, 2022), and this study adopts the same assumption, applying it to the upper half of the building's floors. Various studies report different percentages of water used for toilet flushing, ranging from 10 % to 30 % (Cook, Sharma & Gurung, 2014; Carragher, Stewart & Beal, 2012; Schuetze & Santiago-Fandiño, 2013; Cole & Stewart, 2013). This study adopts Cook et al.'s assumption

of 12 % for toilet flushing in Australia. Fig. 5 shows the water demand per capita in 30-minute intervals for toilet use in Melbourne. Finally, multiplying the temporal coefficient factor by the number of water users on the upper half of the building's floors provides the water demand at each 30-minute interval for different buildings.

3.3. Optimisation algorithm

Metaheuristic algorithms have gained popularity due to their effectiveness in addressing complex problems that are nonlinear, non-convex, and discrete. Consequently, a GA has been employed to determine an optimal operational schedule, improve system profitability, and support the decarbonisation of the electricity supply. The parameters used for the GA are detailed in Proper tuning of these parameters is crucial for achieving optimal performance, as they greatly influence the probability of discovering near-optimal solutions within a feasible amount of computational time. The parameters listed in Table 4 were fine-tuned through sensitivity analysis using a fixed random seed. Proper tuning of these parameters is crucial for achieving optimal performance, as they greatly influence the probability of discovering near-optimal solutions within a feasible amount of computational time.

To guarantee convergence of the optimisation runs, the hypervolume indicator (HI) has been computed using random seeds from 1 to 1000. The HI is especially suitable if the optimal front is indeterminate. It evaluates the area between the optimal solutions and a predetermined reference point, which must be higher than the maximum values on the Pareto front (Zitzler et al., 2003). To address the stochastic nature of the GA and increase the reliability of the results, the GA is executed multiple times with different random seeds. For each case, the GA runs are repeated five times in this study to ensure robust outcomes.

3.4. Stakeholders

A pivotal question is who can benefit from the output of this research? The two primary stakeholders are owners, corporations and energy retailer companies. Owners' corporations manage the common property of residential, commercial, retail, industrial, or mixed-use

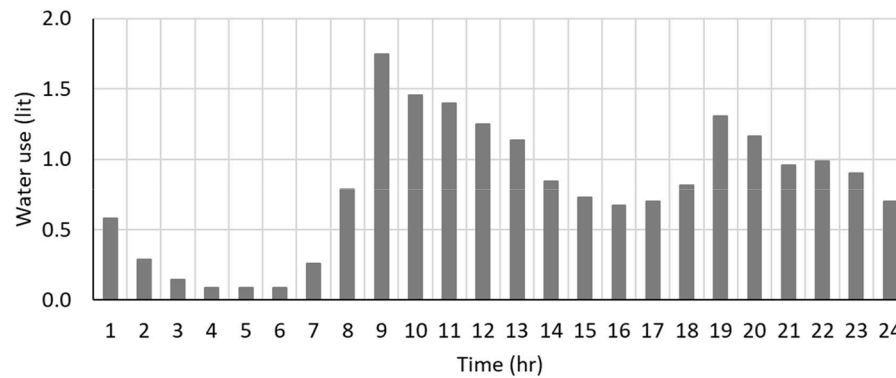


Fig. 5. Diurnal water consumption profile per capita for toilet flush in residential buildings in Melbourne.

Table 4
GA parameter and inputs.

GA attributes		Value
Algorithm type Parameters	Population	Rank and Crowding Survival GA
	No. offspring	480
	No. Evaluations	50
	Crossover probability	220,000
	eta	0.6
	Mutation probability	15
	eta	0.1
		20

property developments. For these stakeholders, implementing a MPS system in high-rise buildings offers significant financial benefits and provides credits for achieving sustainability certifications. Customers can access to real-time wholesale electricity prices in the Australian energy market through third-party companies such as Amber Electric. Unlike traditional energy retailers, which mark up the wholesale cost of electricity, Amber passes these prices directly to owner's corporations for a flat monthly subscription fee. This approach not only enables owners' corporations to earn through energy arbitrage but also supports the transition to renewable energy sources in Australia. Additionally, integrating such a system enhances the building's sustainability profile, potentially providing credits in Green Star and NABERS ratings by decarbonising the supplied electricity, saving energy in the water distribution network if water is sourced either from a nearby stream or rainwater collection system, and enhancing liveability through green roofs.

Energy retailer companies can also gain significant advantages by installing MPS systems in high-rise buildings and paying rent to owner's corporations for the utilised space, particularly in targeted suburbs. These systems help alleviate grid bottlenecks by offering decentralised energy storage, thereby smoothing out demand peaks and enhancing grid stability. By reducing the strain on the grid and facilitating more efficient energy distribution, MPS systems promote the transition to greener energy solutions and contribute to overall grid resilience and reliability. Additionally, retailers can benefit by obtaining carbon credits and trading these credits, further supporting their sustainability goals. This alignment between the investment by owners' corporations and energy retailers and the direct financial, sustainability, and operational benefits ensures that there is minimal risk of mismatch between who pays for the MPS system and who gains from its deployment.

4. Result and discussion

This section presents a summary of the key findings derived from the applied methodology. Firstly, the monthly operation Pareto front is discussed using a representative case from the assessments. Next, the impact of operation optimisation on annual arbitrage and emissions is

analysed for both ES and NPWS modes, compared to the base case. Additionally, a WEN analysis is conducted for cases in NPWS mode, illustrating its effect on the economic feasibility and emission of the MPS system. Finally, the impact of the MPS on the building's sustainability rating is evaluated.

4.1. Impact of optimised operation schedule on es and NPWS modes

The developed framework optimises the operation schedule for defined cases on a monthly basis to minimise operational emissions and maximise energy arbitrage. Fig. 6 presents the generated Pareto fronts for each month for the T/T-300 configuration, with other configurations exhibiting similar behaviour. In Fig. 6-a, each dot represents a distinct schedule. To facilitate a reasonable comparison with the base case, which assumes ideal operation for maximum energy arbitrage, the dot with the highest energy arbitrage value is selected for each month. These dots are highlighted with red circles in Fig. 6-a. For all assessed cases in this study, the monthly schedule representing the highest energy arbitrage is chosen. Because almost the economic aspect tends to outweigh the environmental one when making decisions about financial investments in environmental projects (Gordova et al., 2022). The annual operational emissions and arbitrage for each case are calculated through the summation of monthly values for optimal schedules with the highest energy arbitrage. The upward concavity observed in the Pareto fronts suggests a diminishing return in the trade-off between minimising emissions and maximising arbitrage. As emissions are reduced further, each additional reduction results in progressively smaller gains in arbitrage. Conversely, as the focus shifts toward maximising arbitrage, the potential for reducing emissions decreases significantly, leading to higher emissions for marginal increases in arbitrage. This characteristic curve indicates that while initial trade-offs between these conflicting objectives may be efficient, achieving very low emissions alongside high arbitrage becomes increasingly challenging and less efficient, reflecting the inherent conflict between these two objectives. These trade-offs between emissions reduction and arbitrage align with findings by Aryai and Goldsworthy (Aryai & Goldsworthy, 2022), who showed that prioritising emission reductions in storage systems may compromise financial returns in current Australian market conditions. However, unlike prior work that focuses solely on batteries, our study incorporates dual-function MPS systems, demonstrating that integrating water reuse creates new pathways for viable decarbonisation.

To explain the discontinuities in the Pareto fronts, Fig. 6-b illustrates the schedule for June. It is evident that as the cumulative daily operating hours (considering the total pump and turbine operational hours) decrease, a discontinuity appears in the Pareto front. This step-change behaviour indicates that reducing the system's operating hours leads to a significant reduction in revenue, while the change in emissions remains negligible.

Optimising the operational schedule on a monthly basis resulted in a

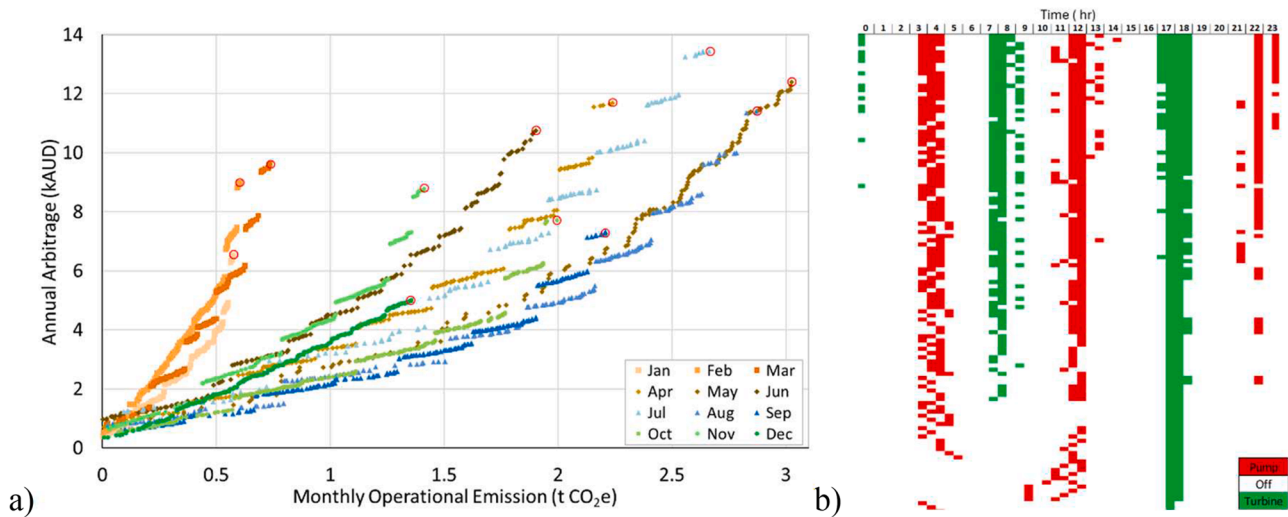


Fig. 6. Optimal monthly operation of the T/T-300 configuration under ES mode; a) Pareto fronts; b) representative operational schedule for the Pareto front in June.

reduction in annual arbitrage compared to the daily operation plan defined in the base case scenario. For ES mode, this reduction is relatively small, varying from A\$114 per year to A\$739 per year depending on the system configuration (Fig. 7). Although the arbitrage reduction ranges from <1 % up to 8 %, the optimisation's impact on emission reduction is significantly greater, varying from 4 % to 25 % depending on the system configuration (Fig. 8). A comparison between Figs. 7 and 8 demonstrates that the small reduction in arbitrage due to the operation of MPS with an optimised monthly schedule leads to noticeable emission reductions in ES mode, especially for configurations using a nearby stream as the LS.

The decrease in arbitrage is more substantial in the NPWS operational mode than in ES mode, and this reduction ranges vary from about 20 % to 30 % for different configurations. Generally, as the building height increases, the difference in arbitrage between the base case and NPWS rises. However, this decrease in arbitrage results in significant emission reductions, ranging from 25 % to 43 %. In cases where a nearby stream is used as the LS, supplying part of the non-potable demand through the stream reduces the water demand from the urban water distribution network, leading to energy savings in the urban water supply network. Considering the WEN in the water network increases emission reduction from 43 % to 52 %, as shown in Figs. 7 and 8. Hence,

from an environmental perspective, NPWS mode prevails over ES mode.

To better understand the financial impact of targeting operational emission reductions on MPS, the cost per ton of CO₂e reduction for the 17 configurations under both the ES and NPWS modes are calculated by dividing the arbitrage loss by the emissions reduced and summarised in Table 5. As shown in Table 5, the cost of decarbonisation through an optimal monthly schedule ranges from A\$20 to A\$226 per ton of CO₂ for ES mode and from A\$72 to A\$229 per ton of CO₂ for NPWS mode. Given the current value of carbon credits in Australia, approximately A\$25, decarbonising stored energy via MPS may not be attractive for stakeholders under most configurations unless carbon tax policies and carbon credit values are adjusted. It is also important to note that these calculations do not account for the revenue earned from water bill savings, only the changes in energy arbitrage. When water bill savings are considered, decarbonisation can become a very economically attractive scenario, as discussed in Section 4.2.

4.2. Assessment of NPWS mode through the WEN lens

By incorporating the operating status of NPWS for the top half floors in residential buildings (gravity-fed through stored water in the US), the operating hours and consequently emissions of the MPS system are

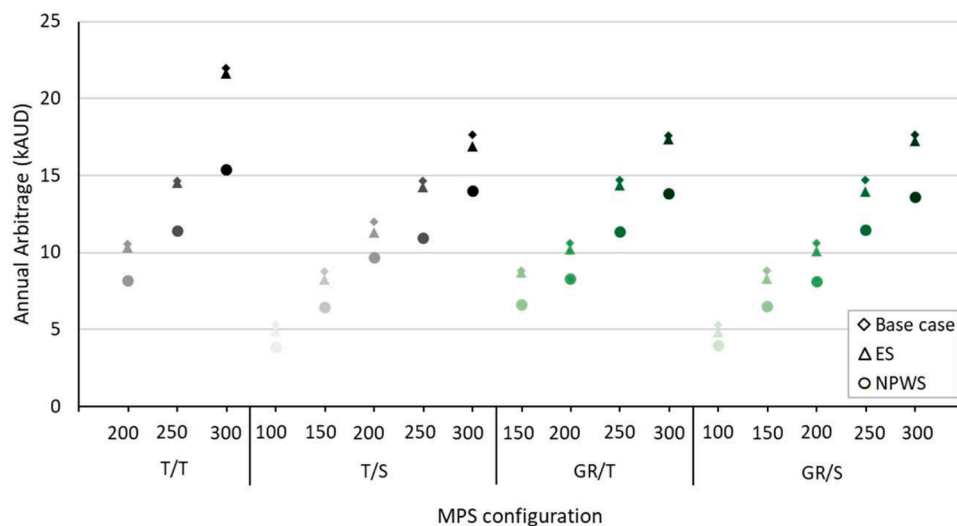


Fig. 7. Annual arbitrage of different MPS configurations under three operational modes.

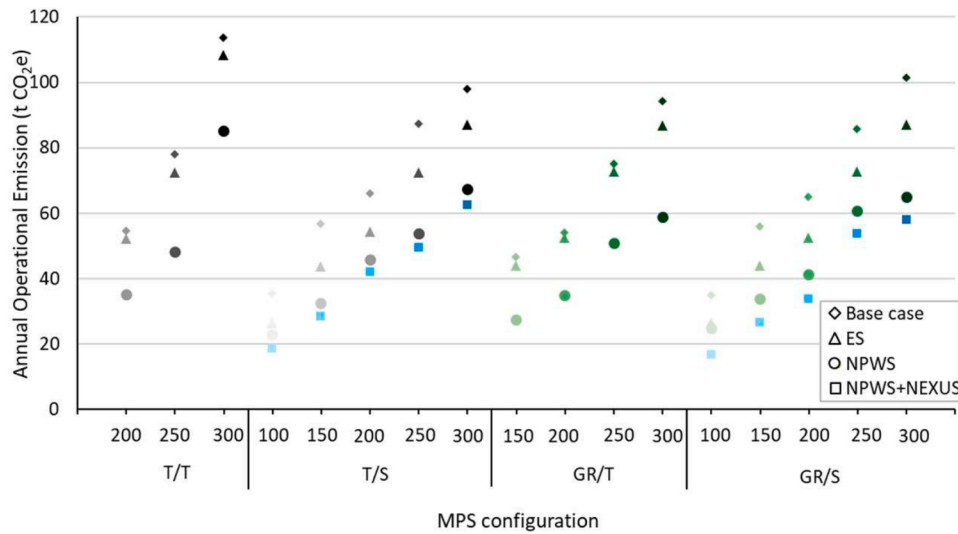


Fig. 8. Annual OE of different MPS configurations under three operational modes.

Table 5

Decarbonisation cost (AUD/ton CO₂e) for different configurations under ES and NPWS operational modes.

MPS configuration	ES	NPWS
TT-200	109	122
TT-250	20	109
TT-300	66	229
TS-100	42	86
TS-150	38	82
TS-200	62	98
TS-250	27	98
TS-300	68	104
GRT-150	49	114
GRT-200	226	120
GRT-250	148	138
GRT-300	36	106
GRS-100	52	72
GRS-150	42	79
GRS-200	37	79
GRS-250	56	102
GRS-300	29	94

reduced across all MPS configurations. Although more operating hours can yield greater energy arbitrage benefits, the algorithm selects the water supply status at certain times to reduce emissions. Fig. 9 illustrates the optimal schedule under ES mode versus NPWS mode for the GR/S-300 configuration, serving as a representative example, while all other configurations exhibit similar behaviour.

The optimal schedule of the GR/S-300 indicates that the water supply mode is not proposed from 7 AM to 1 PM, and it is scattered throughout the rest of the day, primarily from 9 PM to 6 AM. This pattern is influenced by the electricity spot price and emission factor data. Under typical conditions, the pump operates whenever there is a water demand, leading to emissions based on the emission factor at the time of demand. However, in NPWS mode, the energy consumption load of the building's water supply system is managed by storing water in the US, based on the spot emission factor and electricity spot price. Thus, NPWS mode also offers a demand management opportunity for the building's water supply system.

Configurations that utilise a nearby stream to operate the MPS and meet part of the non-potable water demand result in significant water savings in the water distribution network, consequently reducing both

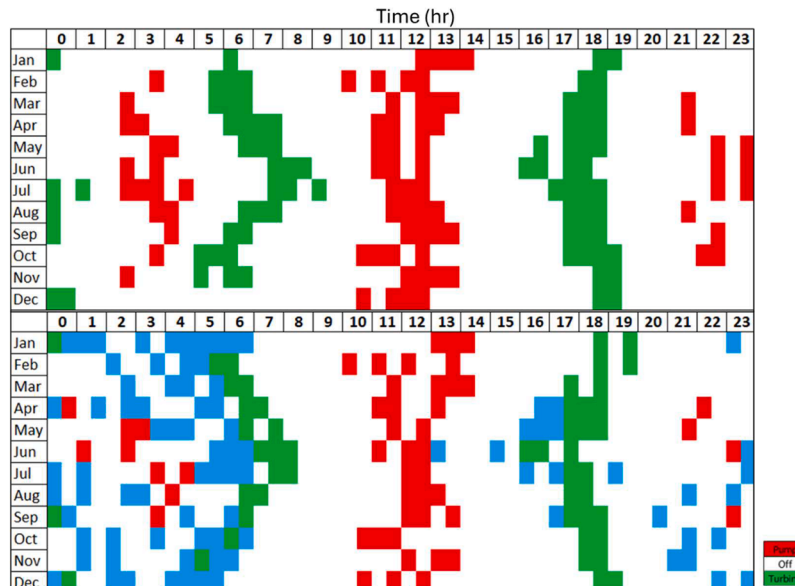


Fig. 9. Optimal operation schedule of the GR/S-300 configuration for ES mode versus NPWS mode.

energy consumption and emissions. A real-world example of a similar initiative is the irrigation system at the University of Melbourne's Parkville campus, partially supplied by withdrawing stormwater from an underground creek passing through the campus. This system has led to substantial savings in the university's water bills. This example demonstrates the feasibility of withdrawing water or stormwater from existing streams.

In this study, the reduction in potable water uses and the resultant energy savings in the water distribution network are calculated for the optimal schedule of configurations with a nearby stream. As summarised in Table 6, the optimal schedule can reduce potable water consumption by approximately 1800 to 3000 m³ in a single building annually. Considering the energy intensity of the water distribution network in Melbourne, individual buildings applying various MPS configurations bring an energy saving of approximately 3 to 5 MWh per year and a reduction of 2.5 to 4 tons of CO₂ in urban water distribution network. While this study adopts fixed input assumptions for water demand, building configuration, and energy intensities, these parameters can vary between projects. A formal sensitivity analysis is beyond the scope of this work but is essential for future research. In particular, building height, non-potable water demand, stream proximity, and the carbon intensity of water distribution should be prioritised for parametric testing, as they directly affect system feasibility and the magnitude of WEN benefits.

The detailed results of the WEN analysis at urban scale are presented in Table 7. Given the number of buildings suitable for MPS installation (Boroomandnia et al., 2024), the deployment of MPS systems across 12 candidate buildings with TS configurations and 11 candidate buildings with GRS configurations would collectively save over 56,500 m³ of potable water annually. At the community scale, the potential energy savings are approximately 54 MWh/year for TS configurations and 44 MWh/year for GRS configurations. This would result in annual CO₂ emission reductions of 44 and 36 tons, respectively, in the water distribution network by implementing TS and GRS configurations across Melbourne. These findings demonstrate the substantial potential for environmental benefits at the urban scale through widespread MPS adoption, reinforcing the role of integrated water-energy strategies in supporting citywide decarbonisation efforts. The urban-scale WEN analysis highlights the broader potential of deploying MPS systems across suitable high-rise buildings. However, the scalability of this approach depends on several practical factors. These include building-specific constraints such as rooftop load-bearing capacity, available roof area, and access to suitable lower reservoirs like streams. In Melbourne, such conditions were met for only a subset of the building stock, as shown by the filtered dataset. Nonetheless, the optimisation model developed in this study is transferable and can be applied to other cities with similar urban forms. This creates an opportunity for precinct-scale planning and water-energy synergy strategies to support broader decarbonisation and sustainability goals.

From a decision-making perspective, the developed optimisation

framework serves as a practical decision-support tool for stakeholders such as owner corporations and energy retailers. It enables users to assess trade-offs under real-world constraints and prioritise scenarios based on economic, environmental, or certification outcomes. This aligns with recent studies (Valencia, Hossain & Chang, 2022) that emphasise the role of integrated infrastructure modelling in guiding urban decarbonisation strategies.

Besides the environmental benefits, the NPWS mode is highly attractive from an economic perspective. This strategy significantly reduces the water bill, saving between 7 and 12 k AUD per year, which noticeably impacts the payback period. Fig. 10 illustrates the economic attractiveness of NPWS, encompassing the WEN approach, compared to other MPS operational modes and Li-ion batteries with equivalent energy storage capacities for benchmarking purposes. Savings in the water bill decrease the payback period in T/S and GR/S configurations by up to 11 and 20 years, respectively. Indeed, savings in the water bill are a game changer, making the NPWS mode the most attractive operating mode, as shown in Fig. 10. In contrast, the changes in the payback period for ES mode compared to the base case operation are negligible. For T/T and GR/T configurations, the payback period varies from 8 to 13 years and 11 to 24 years, respectively, depending on the building height and operating mode. Notably, MPS systems outperform Li-ion battery systems in nearly all configurations, offering shorter payback periods when both energy and water benefits are considered (Boroomandnia et al., 2024). This benchmarking further highlights the economic competitiveness of MPS as a dual-purpose urban infrastructure solution.

4.3. Effect of MPS on the sustainability rating of buildings

Green Star Buildings Version 1.C is Australia's leading sustainability certification system for new and established buildings, administered by the Green Building Council of Australia (GBCA) (Australia, G.B.C.O., 2023). It evaluates sustainability performance across eight categories: Responsible, Healthy, Resilient, Positive, Places, People, Nature, and Leadership. Projects accumulate points by meeting the criteria of individual credits, with a total of 100 core points available. Certification ratings are awarded based on the percentage of points achieved: 4 Star (Best Practice) requires at least 15 points, 5 Star (Australian Excellence) requires at least 35 points, and 6 Star (World Leadership) requires at least 70 points. Additional points earned under the Leadership category, such as through Market Transformation claims, can further enhance a project's score beyond the 100 core points.

Within this framework, the implementation of MPS systems in high-rise could contribute up to 24 points across various Green Star credits. Due to its innovative nature, the MPS setup can earn up to 5 points under the 'Market Transformation' credit, which recognises pioneering initiatives that drive the sustainability agenda forward. As a novel solution for urban water and energy management, the MPS system exemplifies leadership in sustainable building practices. Furthermore, the MPS system contributes to emission reduction through effective demand management, aligning with the objectives of the 'Grid Resilience' credit. By optimising energy use and leveraging off-peak electricity for pumping operations, it reduces the building's overall carbon footprint, earning up to 3 points. In addition to its energy and emission benefits, the MPS system significantly reduces potable water demand by utilising reclaimed water resources in NPWS mode and efficiently managing water resources within the building. This aligns with the 'Water Use' credit, earning up to 6 points by conserving a critical resource and reducing the strain on municipal water supplies. By reducing water demand and contributing to peak-load shaving, MPS can also gain 1 point under the 'Climate Change Resilience' credit, as it enables buildings to respond to the direct and indirect impacts of climate change. 'Operation Resilience' is another credit where up to 2 points are achievable, as MPS enhances the building's survivability in the event of critical infrastructure failure (power and water). If implemented with a green roof configuration, 1 point can be earned under the 'Heat Resilience' credit

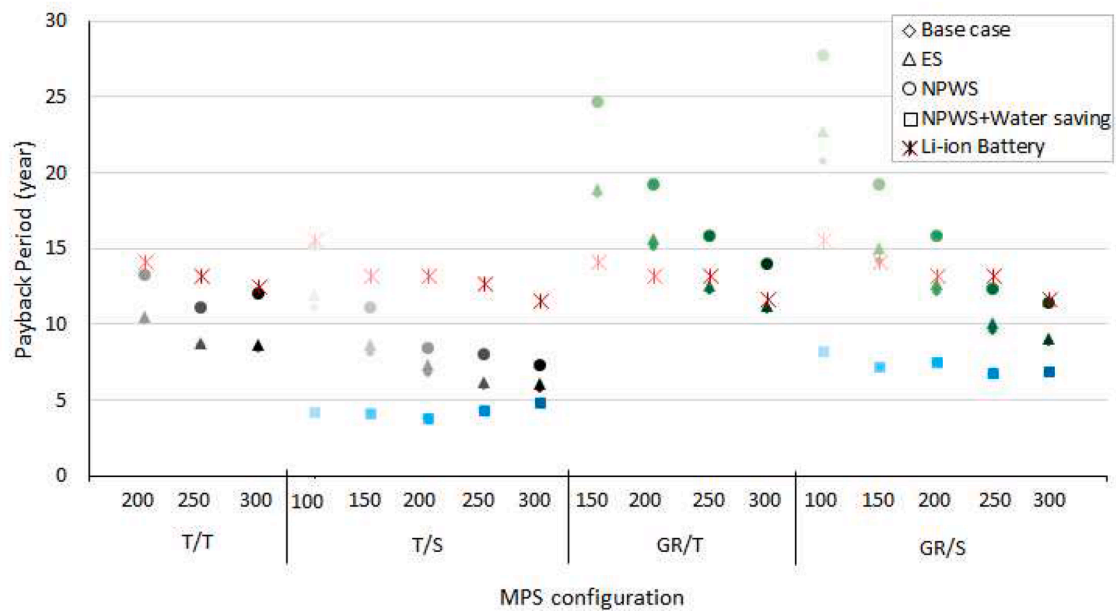
Table 6
Water-energy nexus analysis.

MPS configuration	Annual saved potable water (m ³)	Saved energy in water network (kWh/year)	Extra emission reduction (ton CO ₂)	Annual savings in water bill (k AUD)
TS-100	2542	4424	3.6	10.3
TS-150	2713	4721	3.8	11.0
TS-200	2945	5124	4.2	11.9
TS-250	2372	4128	3.3	9.6
TS-300	1804	3138	2.5	7.3
GRS-100	2320	4037	3.3	9.4
GRS-150	2692	4684	3.8	10.9
GRS-200	2217	3857	3.1	9.0
GRS-250	2309	4018	3.3	9.3
GRS-300	2201	3829	3.1	8.9

Table 7

Water-energy nexus analysis at urban scale for Melbourne.

	Number of candidate buildings (Boroomandnia et al., 2024)	Annual saved water (m ³)	Saved energy in water network (MWh/year)	Extra emission reduction (ton CO ₂)
TS-100	9	22,882	40	32
TS-200	2	5889	10	8
TS-250	1	2372	4	3
Total	12	31,143	54	44
GRS-100	9	20,880	36	29
GRS-200	1	2217	4	3
GRS-250	1	2309	4	3
Total	11	25,406	44	36

**Fig. 10.** Payback period of different MPS configurations and equivalent-capacity Li-ion batteries across different scenarios.

since a portion of the site area comprises green roofing, which reduces the heat island effect. The green roof setup can further increase points under the 'Connection to Nature' and 'Biodiversity Enhancement' credits, earning 2 and 4 points respectively, as it provides a natural space for occupants to engage with.

In summary, the innovative MPS system not only advances sustainability leadership but also delivers measurable benefits in emission reduction, water conservation, resilience, and connectivity to nature. Accelerating the achievement of 24 points through the integration of MPS represents a substantial contribution to the overall certification outcome. It is also worth noting that a Green Star certification increases the building's attractiveness in the real estate market and boosts its value.

4.4. Challenges to adoption

While MPS holds promise as an urban energy storage method, implementing it in high-rise buildings in Australia faces challenges. Structurally, high-rise buildings require thorough planning to accommodate rooftop reservoirs such as modular tanks or green/blue roofs. Although feasible, further cost analysis and upfront carbon assessments are essential to evaluate the financial and environmental impacts, especially for retrofitting existing buildings. Regulatory frameworks also lack explicit provisions for MPS applications, which could lead to

approval delays and inconsistencies in implementation. Additionally, energy arbitrage opportunities are limited because few retailers currently offer real-time access to wholesale tariffs, reducing number of prosumers who could participate in energy trading.

To address these challenges, proposed solutions include developing modular, lightweight reservoir designs to minimize structural loads, establishing specific regulatory guidelines for MPS installations, and increasing the number of retailers offering real-time wholesale pricing to support energy arbitrage. Government incentives could further accelerate MPS adoption by offsetting initial costs and improving the economic feasibility of MPS installations. Potential incentives might include grants or subsidies for energy-efficient building upgrades and tax credits for businesses and property owners who install MPS systems. A government-led pilot installation, tested in a 'real-world' setting and publicized through media, could also facilitate MPS implementation by demonstrating its practical benefits. Additionally, implementing a carbon tax or offering carbon credits for energy storage projects like MPS would motivate adoption by linking it to broader emissions reduction goals. Government-backed funding programs or low-interest loans could also alleviate the financial burden of installations, especially for retrofits. By streamlining regulatory pathways and providing economic incentives, the government could play a crucial role in advancing MPS adoption, supporting Australia's sustainability targets, and promoting innovative energy solutions in urban environments.

Beyond the Australian context, the uptake of MPS systems in other cities and countries may face additional challenges. Differences in electricity market structures, such as the lack of dynamic or wholesale tariffs, could limit the financial viability of energy arbitrage strategies. Climatic factors, including water scarcity or extreme weather conditions, may also constrain the integration of rooftop reservoirs or alternative water sources. Regulatory frameworks vary significantly across regions without clear policies supporting distributed energy storage may encounter approval and operational barriers. Nevertheless, cities with progressive decarbonisation targets, flexible energy markets, and incentives for distributed energy solutions present promising opportunities for wider adoption of MPS technologies.

5. Future outlook

While this study provides valuable insights into the integrated optimisation of MPS systems for operational emissions reduction, economic viability, and contributions to building sustainability certification, several limitations must be acknowledged. First, the analysis was limited to high-rise buildings in Australia, which may constrain generalisability to regions with differing energy markets, climates, or regulatory contexts. Second, the environmental assessment focused exclusively on CO₂ emissions and did not consider other sustainability factors such as water resource use, material lifecycles, or ecosystem impacts associated with water sourcing. Finally, the study did not deeply examine stakeholder perceptions—such as those of building owners, residents, or policy-makers—which are likely to influence the uptake and success of MPS systems in real-world applications.

Building on these findings, several avenues for future research are recommended:

- **Geographic and contextual transferability:** Applying the optimisation framework to other climates, building typologies, and regulatory settings to evaluate its adaptability and performance under varied market conditions.
- **Enhanced water-energy integration:** Exploring the coupling of MPS with rainwater harvesting, greywater reuse, or urban stormwater systems to maximise water-energy synergies and infrastructure co-benefits.
- **Advanced operational optimisation:** Incorporating real-time control strategies that respond to dynamic electricity tariffs, carbon intensities, and demand response signals to increase flexibility and financial returns.
- **Policy and behavioural analysis:** Investigating the influence of carbon pricing, sustainability incentives, and stakeholder preferences on MPS deployment feasibility and scalability.
- **Urban resilience and microgrids:** Evaluating the role of MPS in supporting renewable energy microgrids, backup storage, and decentralised water supply systems in the context of urban disaster preparedness and climate adaptation.

These future directions will not only improve the technical robustness of MPS optimisation but also strengthen its relevance as a scalable, resilient, and sustainable infrastructure solution for cities.

6. Conclusion

This study makes three key methodological contributions: (i) it develops a dual-mode optimisation framework that integrates both energy storage and non-potable water supply scheduling for urban high-rise buildings; (ii) it incorporates penalty-based constraints into the objective functions to ensure emission-aware optimisation under time-varying carbon intensities; and (iii) it applies a monthly scheduling approach that balances emissions and economic arbitrage across different seasonal profiles. Together, these features enable comprehensive assessment of micro pumped storage systems from both

environmental and financial perspectives. Results demonstrate that monthly optimisation of MPS operations can reduce annual emissions by 4 % to 25 % and 25 % to 43 % in the energy storage (ES) and non-potable water supply (NPWS) modes, respectively, indicating significant environmental benefits from optimised MPS operations. The cost of decarbonising through an optimal monthly schedule varies, ranging from A \$20 to A\$226 per tonne of CO₂ for ES mode and from A\$72 to A\$229 per tonne of CO₂ for NPWS mode. Given the current value of carbon credits in Australia, this study confirms the findings of Aryai and Goldsworthy (Aryai & Goldsworthy, 2022), emphasising that electricity storage units in Australia prioritise revenue over decarbonisation. However, considering the water energy nexus (WEN), decarbonisation can become economically attractive.

The NPWS mode shows substantial potential for enhancing sustainability by leveraging a nearby stream to meet part of the non-potable water demand in residential buildings. This not only reduces potable water consumption up to 3000 m³/year but also results in energy savings of approximately 3 to 5 MWh per year and a reduction of 2.5 to 4 tonnes of CO₂. The WEN approach in NPWS mode increases emission reduction from 43 % to 52 %, proving to be more environmentally favourable compared to the ES mode. From an economic perspective, the NPWS mode offers significant savings on water bills thereby substantially impacting the payback period for MPS installations. These savings make the NPWS mode the most attractive operating mode, reducing the payback period by up to 11 and 20 years for T/S and GR/S configurations, respectively. Moreover, by implementing the MPS system, a high-rise building can gain up to 24 points under the following credits: Market Transformation, Grid Resilience, Water Use, Climate Change Resilience, Operation Resilience, Heat Resilience, Connection to Nature, and Biodiversity Enhancement, significantly enhancing its Green Star rating and overall sustainability profile.

In summary, the findings of this study underscore the potential of MPS systems to contribute to urban sustainability and offer a compelling case for their broader adoption in high-rise buildings. However, the practical implementation of MPS systems faces challenges, including operational difficulties and the need for legal permissions from relevant organisations. These hurdles, when compared to the relatively simpler operation of Li-ion batteries, still pose significant barriers that need to be addressed in the practical deployment pathway of MPS.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to only proofread the manuscript, not for technical content generation, data analysis, or model development. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit authorship contribution statement

Arezo Boroomandnia: Writing – original draft, Resources, Investigation, Conceptualization, Visualization, Methodology, Formal analysis. **Behzad Rismanchi:** Supervision, Funding acquisition, Writing – review & editing, Project administration, Conceptualization. **Wenyan Wu:** Writing – review & editing, Supervision, Validation. **Rhys Anderson:** Validation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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